Yash Agarwal

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EDUCATION

Carnegie Mellon University, School of Computer Science — **Pittsburgh, PA**
Masters in Intelligent Information **Systems** - GPA - 3.9/4.0 — December 2026

- *Coursework - Computer Systems, DL Systems, LLM Systems, Deep Learning, Advanced NLP, Multimodal ML*
- *Advisor - Prof.* ***Tianqi Chen*** *(NVIDIA, creator of TVM, XGBoost) - Machine Learning Compiler*

Indian Institute of Technology (IIT), Kharagpur — **Kharagpur, India**
Bachelors in **Computer Science** and Engineering - GPA 8.65/10 — May 2023

- **All India Rank 250** out of 1.2 million applicants for Joint Entrance Examination (JEE Advanced)
- *Relevant Coursework - Deep Learning, Computer Architecture, Operating Systems, Compilers, Computer Networks*
- *Publication* - YesBut: Evaluating Vision-Language Models — **EMNLP 2024**

WORK EXPERIENCE

Modular Inc — **San Francisco, CA**
AI Performance Engineering Intern — May 2026 - Present

- Accelerated pinned (page-locked, DMA-capable) host-memory allocation by **10×** for tiered **KV-cache offload**, cutting a 1.5TB allocation from 50+ minutes to under 5, unblocking **trillion-parameter MoE** serving on 8-GPU nodes
- Built a **lock-free MPSC** pipeline that faults pages across NUMA-pinned threads at 99.2 GB/s, isolates per-region fault locks via low-level Linux VM **syscalls** to eliminate lock contention, and **pipelines** it with chunked GPU registration

Squarepoint Capital — **Bangalore, India**
Software Developer **C++** — July 2023 - July 2025

- Architected new trading capabilities for FX Options, implementing end-to-end solution in **C++ 20 and Python** using **STL, object oriented design** and comprehensive **tests**, that pioneered new asset class capability at the firm
- Redesigned messaging protocol by refactoring 2000+ line C++ protocol into modular object oriented structure
- Debugged critical production crashes using **gdb** core dump analysis and implemented **thread-safety** in memory access
- Redesigned **multithreaded** Python testing framework to single-threaded architecture eliminating **40%** test flakes

PROJECTS @ CARNEGIE MELLON

Malloc, cache, shell | Computer Systems Course Projects — **Spring 2026**

- Implemented **malloc** using **mmap** and free-lists - implicit, explicit, segregated, block coalescing and various fit policies
- Optimized **throughput** using Better Fit policy and memory **utilization** using **seg-lists** with miniblocks and no footers
- Implemented a **cache** simulator, optimized matrix transpose by tiling and eliminating **evictions**, 4th in a class of 200
- Implemented unix shell with signal handling, bg/fg commands, forking and reaping child processes and IO redirection
- Parallelized a File System with **RW locks**, enabling **concurrent** reads and writes to shared inodes, blocks, directories

Needle - Optimizing CUDA GPU Kernels and Autodiff | DL Systems Course Project — **Fall 2025**

- Optimized Matmul **CUDA kernel** by tiling, vectorization and GMEM coalescing, achieving 85% performance of **cuBLAS**
- Implemented **Flash Attention**, including kernel fusion, tiling, online softmax, achieving 77% performance of PyTorch
- Architected automatic differentiation Tensor, complete with mathematical ops and nn.Modules for Conv, RNN, MHA
- Developed NDArray class supporting reshaping, broadcasting, summation, Matmul using **C++ and CUDA** backend

WhisQer - Quantization trained Conformer | Deep Learning Course Project — **Fall 2025**

- Architected **quantized** Conformer ASR with 2-bit and 1-bit precision variants, with learnable scaling parameters, compressing a 108MB FP32 model to 45.6MB 2-bit by quantizing encoder FFN and attention projections
- Implemented teacher student co-training combining CTC and KL divergence loss and straight through estimator
- Built end-to-end PyTorch ASR training pipeline with streaming audio dataloader, feature pipeline, and CTC decoding

SKILLS

Programming Languages and Tools - C++ 23, Python, Rust, Bash, Git, Gitlab, Linux, Bazel, CMake, STL
Machine Learning - PyTorch, JAX, Numpy, CUDA, Triton, Mojo, Wandb, HuggingFace